

Sentence Memorability Experiment

How Noun Memorability and Grammatical Voice Shape Semantic and Syntactic Memory Retention

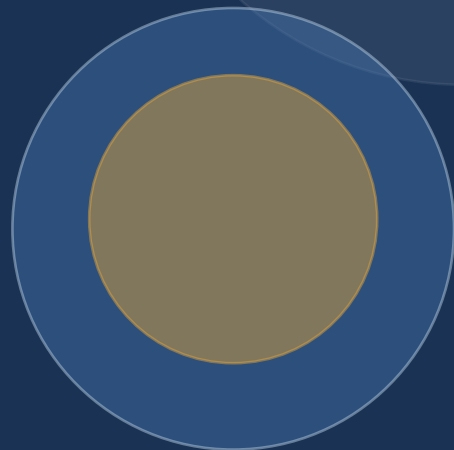
Team: Idle Death Gamblers

Members: Shreyas Deb · Yajat Lakhanpal · Prasoon Dev

Paradigm: Continuous Recognition Memory (Online)

Design: 4 Noun Conditions × 2 Voice (Within-Participants)

N: 112 Participants | 5,264 Encoding Trials



Background & Theoretical Motivation

Central Question: Is sentence memory a faithful archive — or a reconstructive process anchored by meaning?

Kernel-Plus-Tag Hypothesis

Miller (1962), Mehler (1963)

Sentences reduced to a core meaning kernel; syntactic features (e.g., passive voice) stored as forgettable independent tags.

Reconstructive Memory

Bregman & Strasberg (1968)

Rejected the all-or-none tag model. Second guesses were above chance, proving syntactic form is reconstructed from imagery and noun salience.

Gist vs. Wording Dissociation

Begg (1971), Begg & Paivio (1969)

Meaning accuracy declines with lag ($r = -.996$); wording accuracy does not ($r = -.107$). Wording is rebuilt from mental image at retrieval.

Anchoring Reconstruction

James et al. (1973)

Free-recall paradigm with HH/HL/LH/LL noun conditions — salience bias toward subject position; active-sentence preference. → Present study extends this to online continuous recognition.

Research Hypotheses

H1

Gist / Meaning

Corrected Immediate Recognition (Corrected IR) will differ significantly across noun memorability conditions (HH, HL, LH, LL). Sentences with at least one high-memorability noun should be better recognised.

*Operationalised as:
Effect of noun condition
on Corrected IR
(Kruskal-Wallis)*

H2

Verbatim / Wording

Wording Recognition (WR) accuracy will be independent of noun memorability and will NOT be significantly affected by the word-condition manipulation.

*Operationalised as: No
effect of noun condition
on WR Accuracy*

H3

Voice Effect

Active voice sentences will show different memorability / wording recognition patterns compared to passive voice, reflecting reconstructive biases toward active syntax.

*Operationalised as:
Voice × Condition
interaction on both IR
and WR*

Methods: Design & Stimuli

Experimental Design

4 (Noun Condition) × 2 (Voice) — fully within-participants

Code	Subject Noun	Object Noun
HH	High memorability	High memorability
HL	High memorability	Low memorability
LH	Low memorability	High memorability
LL	Low memorability	Low memorability

× **Voice: Active / Passive** → **8 within-participant cells**

S–V–O sentences ≤ 8 words | No proper nouns | Medium-concreteness action verbs
High- & low-memorability nouns from established norms; screened for fluency & coherence

Stimuli & Participants

112

Final Participants

400

Target Sentences (200 active + 200 passive)

50

Sentences per condition per voice

48

Sentences seen per participant (3 blocks)

5,264

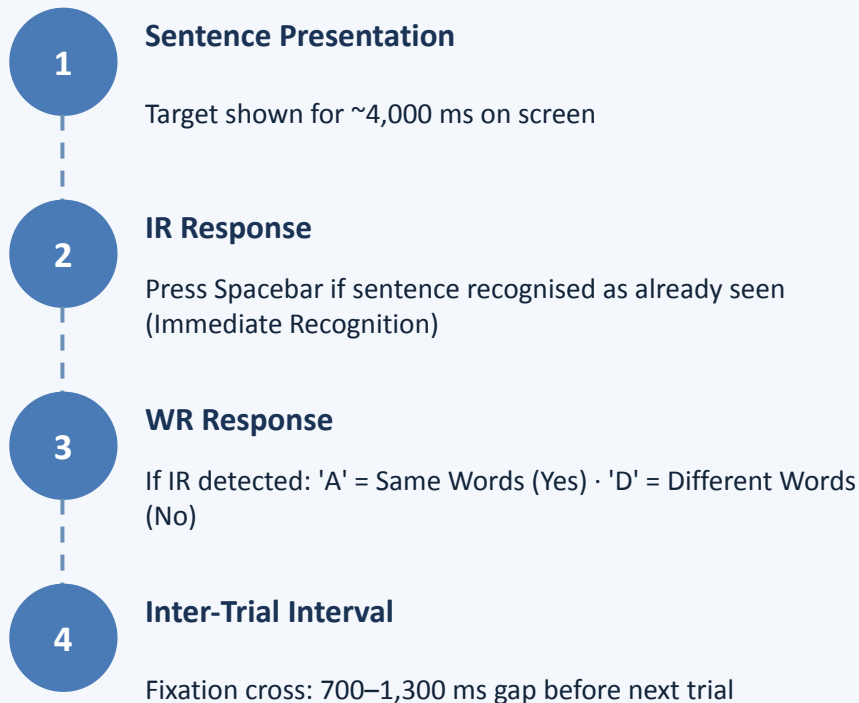
Encoding Trials (perfectly balanced)

96.2%

Block pass rate (329 / 342 blocks)

Procedure & Data Pipeline

Continuous Recognition Procedure



Data Processing Pipeline

Raw Input

81,329 event-log rows across 115 log files

Clean & Filter

Remove practice, gap_time, rest events; retain target trials only

Vigilance Validation

Block pass criterion: $\text{Correct VIRs} > (\text{Wrong IRs} \div 2) + \text{Missed VIRs}$

Feature Extraction

Parse stimulus IDs → noun condition (HH/HL/LH/LL) + voice (A/P)

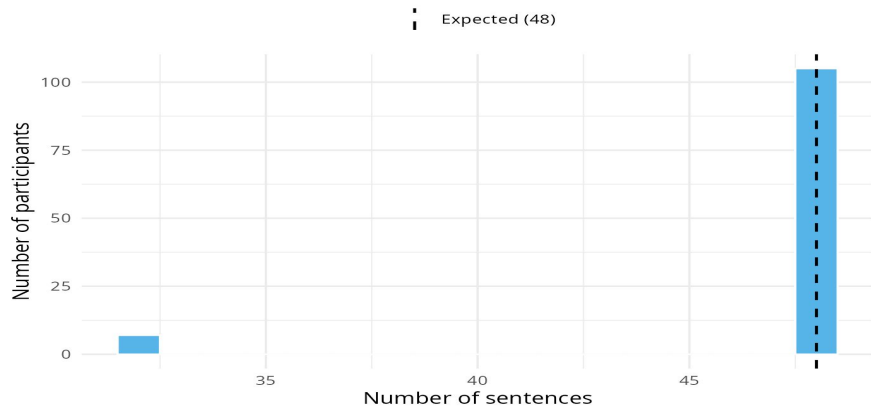
Final Dataset

48,281 observations | 112 participants | Perfectly balanced design

Key Metric: Corrected IR = Hit Rate – Global False Alarm Rate (isolates true recognition discriminability from response bias)

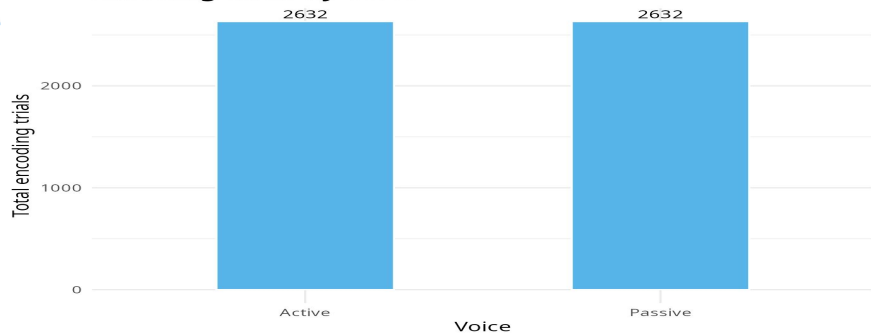
Data Quality & Balance Checks

Target Sentences Shown per Participant



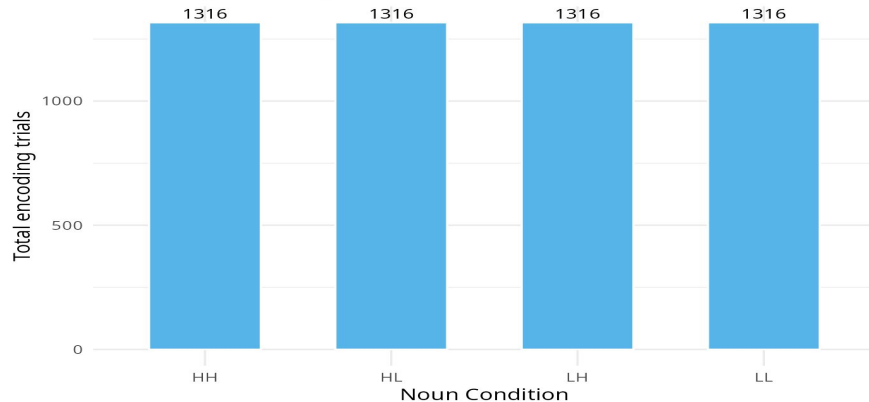
*Sentences per Participant
(most = 48, expected)*

Encoding Trials by Voice



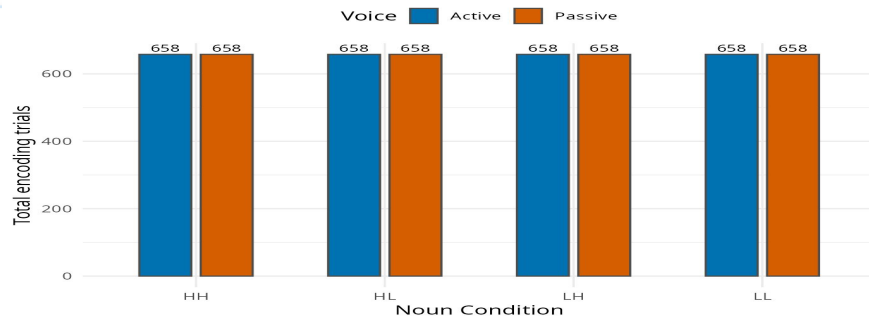
Trials by Voice

Encoding Trials by Noun Condition



*Trials by Noun Condition
(1,316 each – perfectly balanced)*

Encoding Trials: Noun Condition × Voice



Trials per Cell

Statistical Analysis Framework

Normality Assessment (Shapiro-Wilk)

IR Corrected Rate: $W = 0.941, p < .001$ | IR Reaction Time: $W = 0.986, p < .001$ | WR Accuracy: $W = 0.911, p < .001$ → All metrics significantly non-normal → Non-parametric tests justified throughout

Tests Used

Kruskal-Wallis

Main effect of Noun Condition on Corrected IR, WR Accuracy, and IR Reaction Time

*Effect size: ϵ^2
(epsilon-square
d)*

Scheirer-Ray-Hare

Two-way non-parametric equivalent of ANOVA: Noun Condition × Voice interaction

*Simultaneous
main +
interaction
effects*

One-Sample Wilcoxon

Signed-Rank

WR Accuracy significantly above chance (0.5) for each voice condition

*Benchmark
vs. chance*

Wilcoxon Rank-Sum

(Post-hoc, Holm-corrected)

Pairwise comparisons between noun conditions (where Kruskal-Wallis significant)

*Controls
familywise
error rate*

Paired Wilcoxon

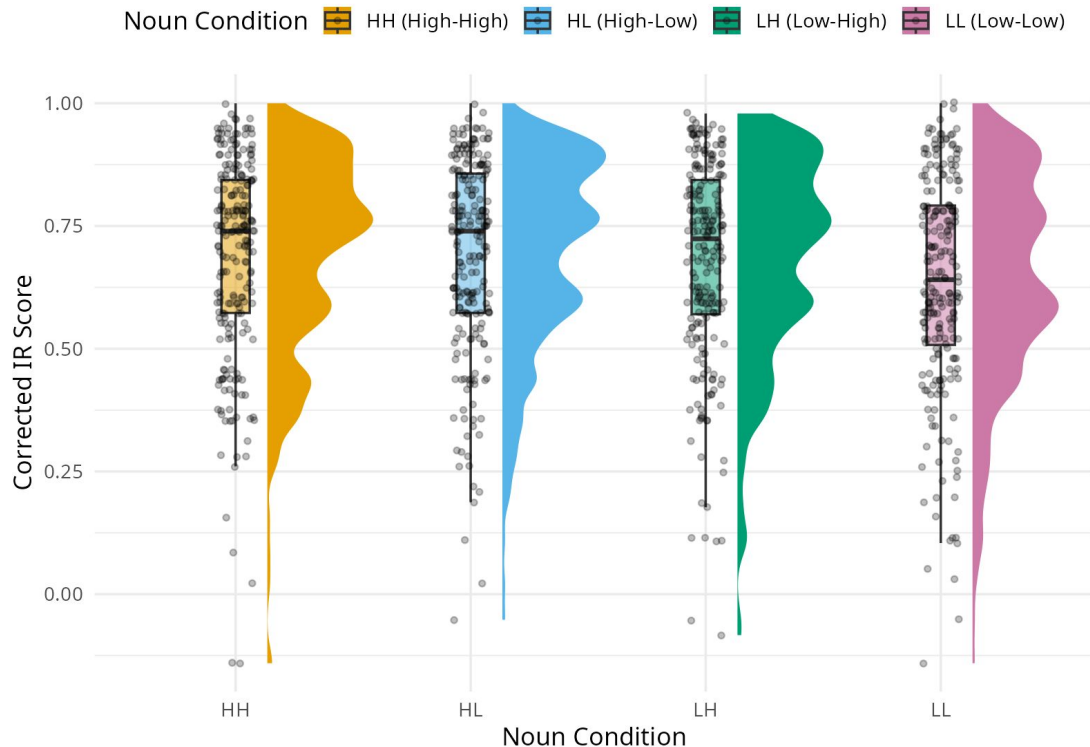
Signed-Rank

Overall voice effect (Active vs. Passive) on both Corrected IR and WR Accuracy

*Within-participant
design*

Results: Corrected IR by Noun Condition (H1)

Corrected IR Scores by Noun Condition



Key Finding

Kruskal-Wallis:

$H(3) = 11.26, p = .010$
 $\epsilon^2 = 0.013$ (small effect)

H1 Supported ✓

- Noun condition significantly affects sentence recognition
- LL (both nouns low) shows the clear deficit
- HH, HL, LH all perform similarly
- One high-memorability noun is sufficient to anchor gist

Descriptive Statistics

Cond	Mean IR CR	SD
HH	0.689	0.210
HL	0.691	0.200
LH	0.679	0.209
LL ↓	0.632	0.224

Post-hoc Analysis: Holm-Corrected Wilcoxon Tests

Pairwise p-values (Holm-corrected)

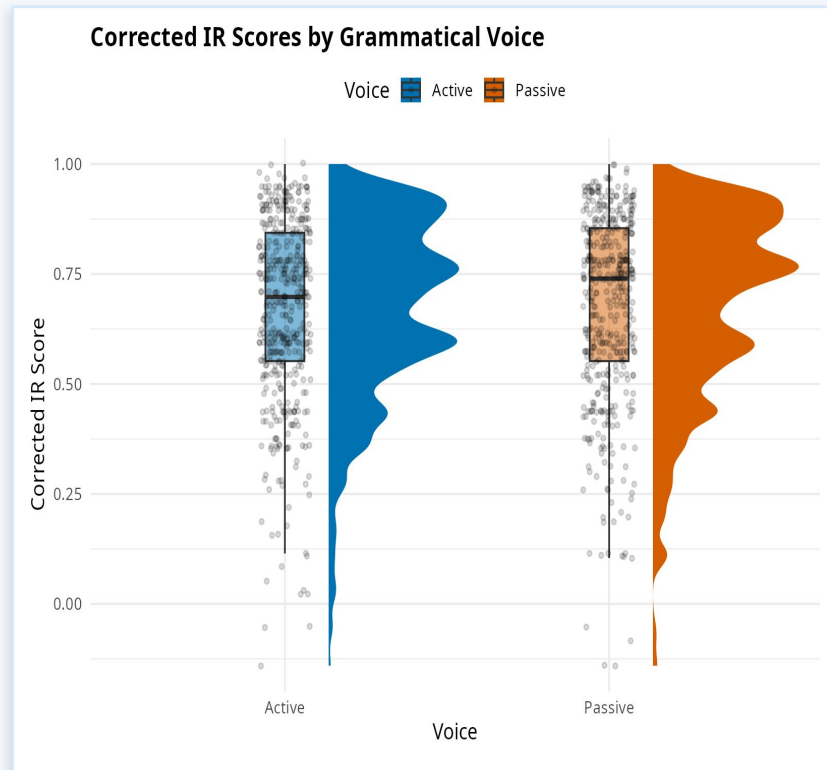
	HH	HL	LH
HL	1.000	—	—
LH	1.000	1.000	—
LL	.024 *	.024 *	.077

* $p < .05$ (Holm-corrected)

Threshold Model of Imageability

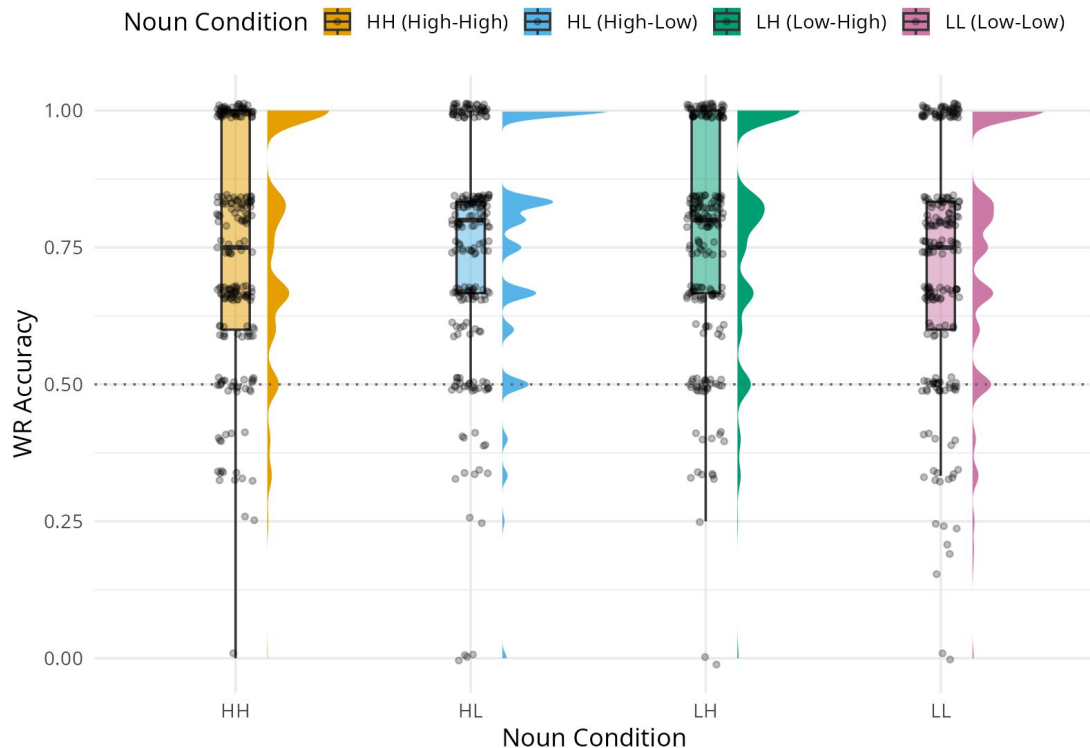
The results reveal a threshold pattern rather than a linear dose-response relationship. The presence of at least one highly memorable noun (HH, HL, or LH) is sufficient to anchor the sentence's gist in memory. Only when BOTH nouns are low-memorability (LL) does recognition suffer. This suggests imagery is anchored by the most salient noun — not averaged across all nouns.

Corrected IR by Voice



Results: Wording Recognition Accuracy (H2)

Wording Recognition Accuracy by Noun Condition



Key Finding

Kruskal-Wallis:

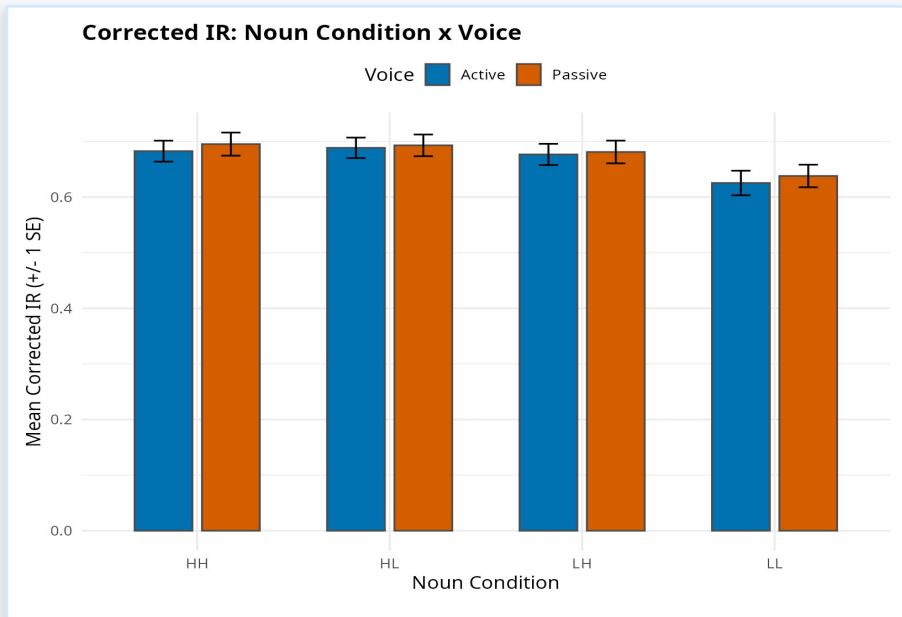
$H(3) = 2.52, p = .471$
 $\epsilon^2 = 0.003$ (negligible)

H2 Supported ✓

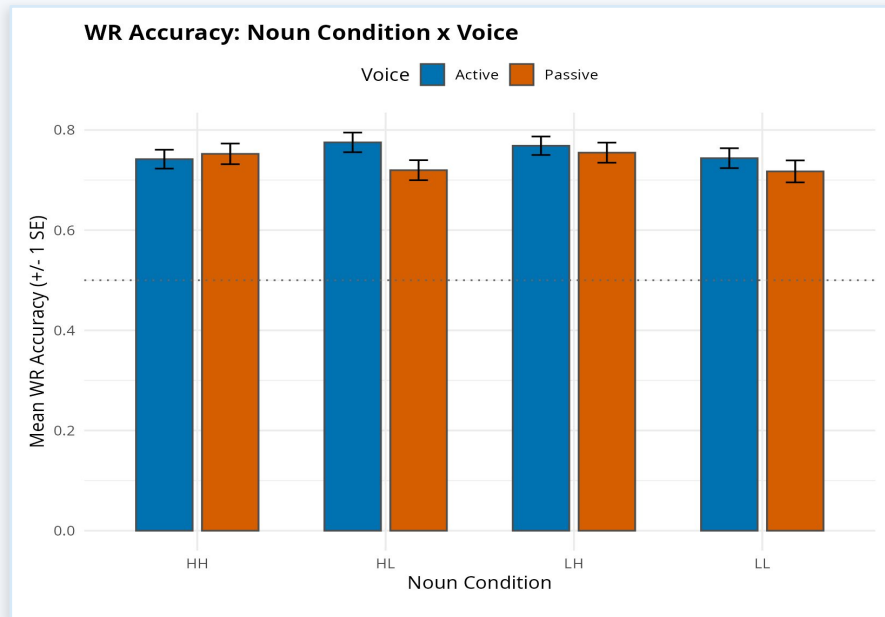
- NO effect of noun condition on WR accuracy
- Performance uniformly high (~73-76%) across all conditions
- WR above chance (0.5) for both Active and Passive (both $p < .001$)
- Confirms: wording memory is independent of semantic imageability
- → Supports gist vs. verbatim dissociation

Results: Noun Condition × Voice Interaction (H3)

Corrected IR: Condition × Voice



WR Accuracy: Condition × Voice



Scheirer-Ray-Hare Test Results:

Corrected IR: Noun condition $p = .010$ ✓ | Voice $p = .384$ ✗ | Interaction $p = .975$ ✗

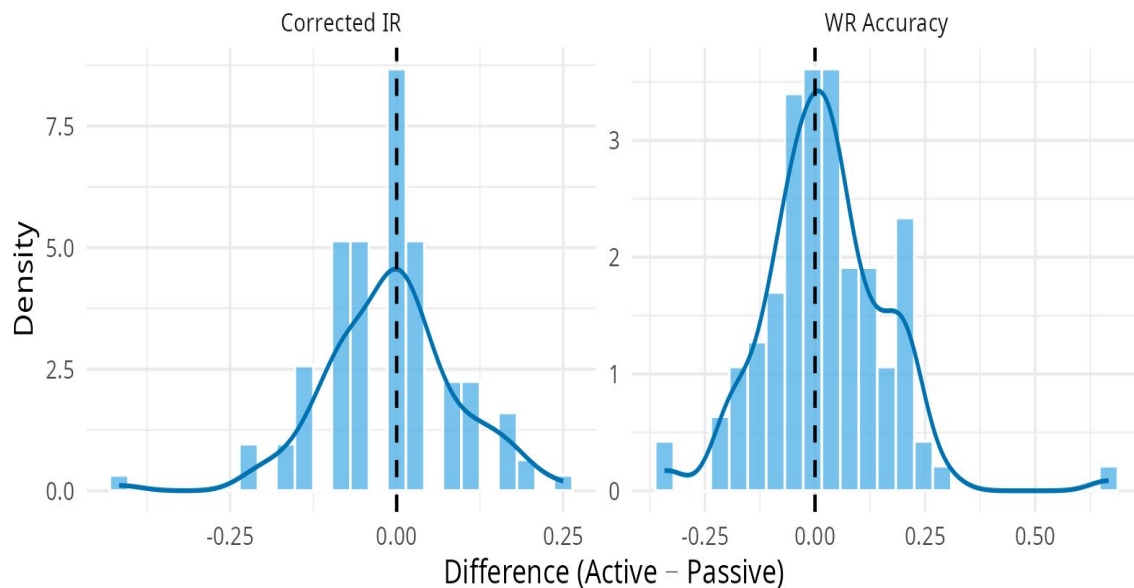
WR Accuracy: All effects non-significant (all $p > .200$) — H3 not supported

Voice Effect: Active vs. Passive Analysis

Within-participant voice differences (Active – Passive)

Voice Effect: Active – Passive (per participant)

Dashed line = zero; centred distribution indicates no systematic voice bias



Paired Wilcoxon Results

Corrected IR

V = 1916.5

p = .485 (n.s.)

Active: M = 0.668 Passive: M = 0.677

WR Accuracy

V = 3520.5

p = .114 (n.s.)

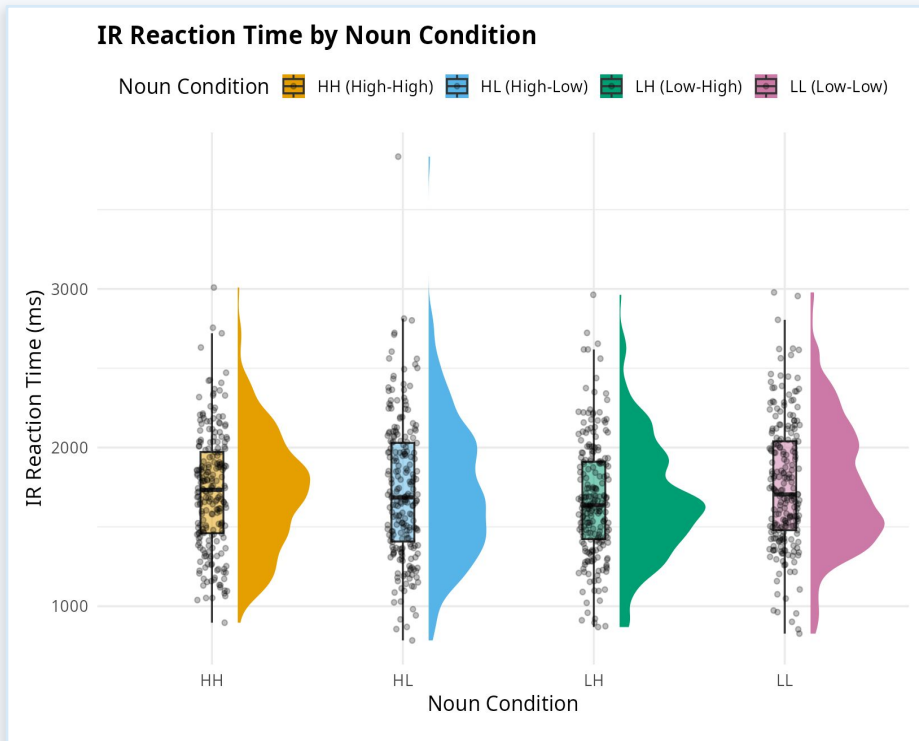
Active: M = 0.756 Passive: M = 0.734

WR slightly favours Active (consistent with James, 1973 active-sentence bias) but does not reach significance. Distributions centred near zero.

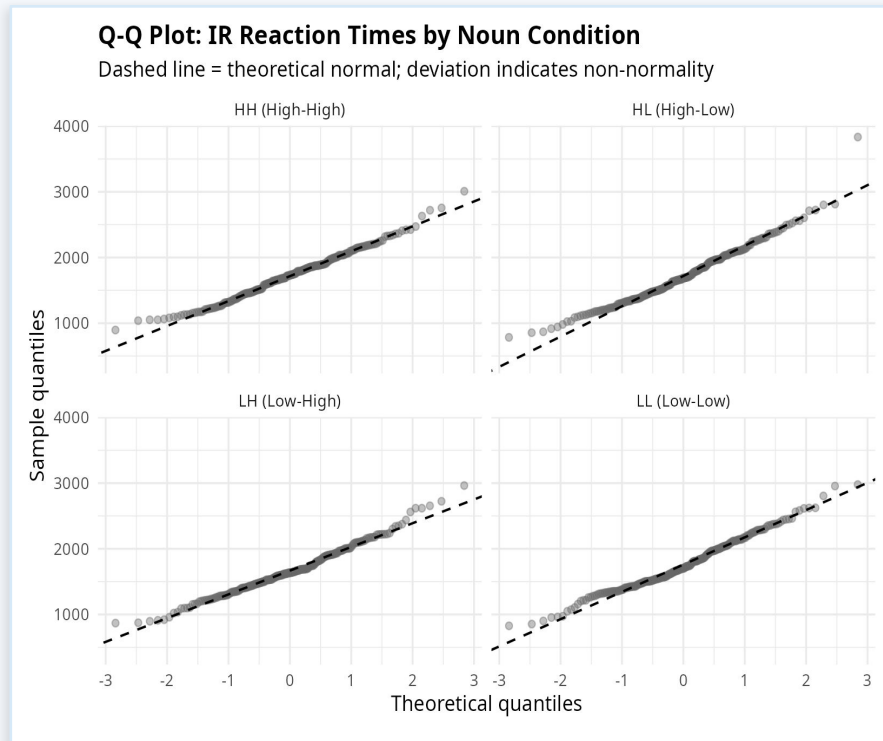
WR Accuracy significantly above chance (0.5) for both Active ($p < .001$) and Passive ($p < .001$) — participants genuinely discriminate voice

Supplementary: Reaction Time & Normality Diagnostics

IR Reaction Time by Noun Condition



Q-Q Plot: IR Reaction Times



Kruskal-Wallis on RT: $H(3) = 6.05$, $p = .109$ (n.s.) | Directional: Passive-LL slowest ($M = 1808$ ms) vs Active-HL fastest ($M = 1664$ ms) — 144 ms spread. Q-Q plots confirm heavy tails justifying non-parametric approach.

Discussion: Threshold Model of Semantic Imageability

Threshold Effect — Single high-memorability noun is sufficient



One vivid noun is enough to create a stable image trace and anchor gist memory

LL is uniquely impaired — when both nouns are abstract, no anchor image can form

Imagery is anchored by the most salient noun, NOT averaged across the sentence

Clean dissociation (IR affected, WR unaffected) matches Begg's (1971) predictions

Core Insight: "It is the imageability of a sentence's constituent nouns — not its grammatical voice — that determines whether its gist survives in memory."

Theoretical Implications



Supports Begg's (1971) Dual-Process Framework

The clean dissociation — noun memorability affects semantic recognition (IR) but NOT wording recognition (WR) — confirms that meaning is stored holistically as an image trace, while surface wording is reconstructed separately at retrieval.



Challenges the Kernel-Plus-Tag Hypothesis

If voice were stored as an independent, fragile syntactic tag (Miller, 1962; Mehler, 1963), passive sentences would decay differently or interact with noun saliency. Instead, robust above-chance WR across all conditions suggests voice is reconstructed — not retrieved — from the semantic trace.



Extends James (1973) to Online Continuous Recognition

James used free recall with HH/HL/LH/LL nouns. Our online within-participant replication at scale confirms the noun-condition hierarchy and extends it to a continuous recognition paradigm — strengthening the ecological validity of the imageability-gist link.



Active-Sentence Bias: Directional but Non-Significant

WR trended toward higher accuracy for Active sentences ($M = 0.756$) vs. Passive ($M = 0.734$), consistent with the active-sentence recall bias documented by James (1973). The pattern did not reach significance — future work with larger N may resolve this.

Conclusions

H1 ✓

Noun memorability significantly affects corrected IR ($p = .010$). LL is uniquely impaired — threshold model confirmed.

H2 ✓

WR accuracy is independent of noun memorability ($p = .471$). Verbatim retention is driven by a separate mechanism.

H3 ✗

No voice main effect or Condition $\times$ Voice interaction on IR or WR. Grammatical voice does not moderate memorability effects.

Future Directions

Multivariate GLM Modeling

Jointly evaluate noun condition, voice, reaction time, and trial position as predictors of recognition accuracy

Signal Detection Theory (SDT)

Compute sensitivity (d') and response bias (c) for a deeper analysis beyond binary accuracy

Practice Effects Analysis

Compare practice-phase vs. main-block performance to evaluate familiarity and learning effects